

INSC-SPEC: Specimen-Level Insect Species Recognition Framework from Multi-View Images and Videos

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Context & Motivation

Context

- Insects are critical to ecosystems, contributing to pollination, nutrient cycling, and pest control.
- Manual species identification is slow and requires rare taxonomic expertise.
- Deep learning provides an automated alternative and is an active research area in the biological research.

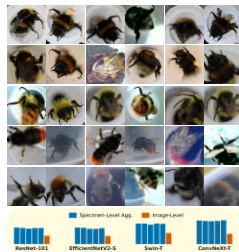


Figure : Multi-view frames of same specimens

Research Gaps

- Most recognition approaches^a treat each image independently; no specimen-wise organisation characterized.
- Most image-level approaches work on high-quality photographs from museum collection / lab specimens.
- Limited work investigates how multiple views and videos of same specimen can be effectively used.

^aSpiesman et al., 2021; Zhang et al., 2025

Broad Objective



(a) *B. Terrestris*



(b) *B. Pascuorum*

Figure : Images from the BumB-Spec Dataset

Objective: Specimen-Level Species Recognition of live insects from an imperfect dataset (limited samples, class imbalance, heterogeneous sources).

Contributions:

- INSC-SPEC: a specimen-level training & evaluation framework for multi-view insect datasets.
- BumB-Spec, a dataset of multi-view photos/videos of live bumblebee specimens; 4 iNat.-derived datasets (ladybird, hoverfly, jewelwing, butterfly).
- Quantitative Evaluation, Ablation Studies, Generalization Studies to show the effective of framework and Specimen-level result aggregation.

INSC-SPEC Overview

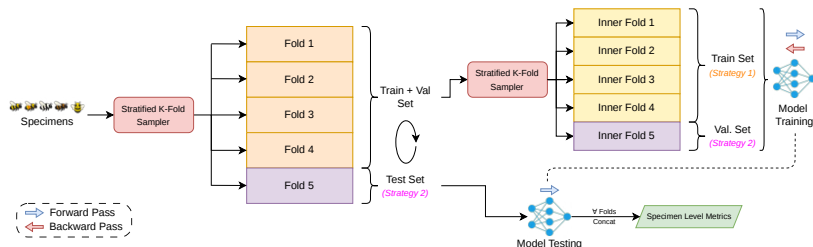


Figure : Overall Framework overview using cross-validation and stratification on the specimens. Strategy 1 and 2 are for frame extraction within a specimen

- **Core idea:** treat the *specimen*, not the image/frame, as the unit of training & evaluation; extract frames from specimens later
- Stratified 5-fold cross-validation on specimens $\Rightarrow$ balance class-to-specimen ratio per fold.

Strategy 1 — Training Set

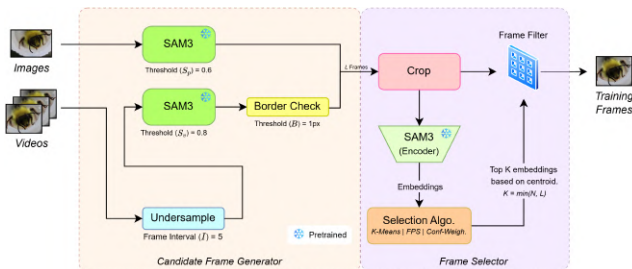


Figure : Strategy 1 for Training Frame Extraction, focusing on balancing number of frame for each specimen.

- **Goal:** Extract a fixed number of diverse information frames on the specimen
- Filter eligible frames $\Rightarrow$ Select top N diverse frames

Strategy 2 — Val and Test Set

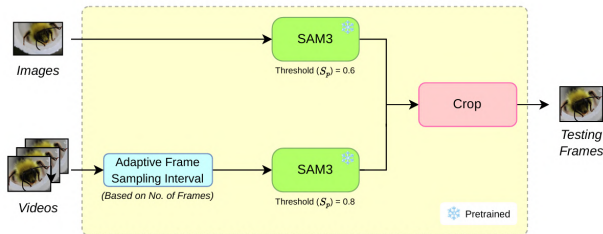


Figure : Strategy 2 for Testing Frame Extraction, focusing on extraction of enough frames for each specimen as possible.

- **Goal:** Extract large number of near-exhaustive views for each specimen.
- Retain all images and adaptively sample video frames based on video length:

$$Interval(N) = \begin{cases} 5, & N < 150, \\ 15, & 150 \leq N < 600, \\ 30, & 600 \leq N < 1500, \\ 60, & N \geq 1500. \end{cases}$$

Datasets Overview

Group	Genus	Source	Media	#Species	#Specimens
Bumblebee	<i>Bombus</i>	Ours	I+V	5	181
Jewelwing	<i>Calopteryx</i>	iNaturalist	I	8	698
Butterfly	<i>Vanessa</i>	iNaturalist	I	8	1098
Ladybird	<i>Coccinella</i>	iNaturalist	I	8	1279
Hoverfly	<i>Eristalis</i>	iNaturalist	I	8	1382

Table : Overview of all datasets.

- *BumB-Spec*: Bumblebee dataset acquired through 3 years of fieldwork + lab recordings (IRBI, Tours); Contains images and videos at specimen-level.
- Generalization Dataset(s): crowdsourced dataset(s) constructed from research-grade observations from iNaturalist; Contains only multi-view images for each specimen.

Note: For all experiments, only BumB-Spec dataset is considered, while other datasets are for generalization experiments only.

BumB-Spec (*Bumblebee*) Dataset

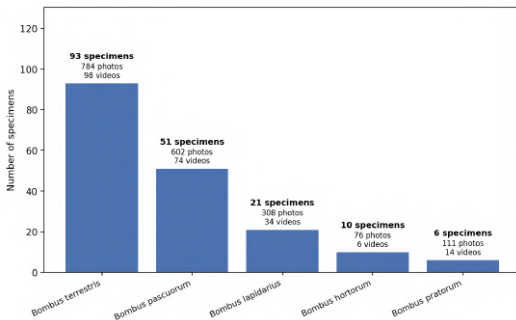


Figure : Distribution of Specimens, Photos, and Videos across BumB-Spec dataset.

Species Characteristics:

- *B. terrestris*: 2 yellow bands, buff tail
- *B. hortorum*: 3 yellow bands, white tail
- *B. pratorum*: reddish-orange tail
- *B. lapidarius*: black body, red tail
- *B. pascuorum*: brownish-orange coloration

BumB-Spec: Media Distribution in Specimens

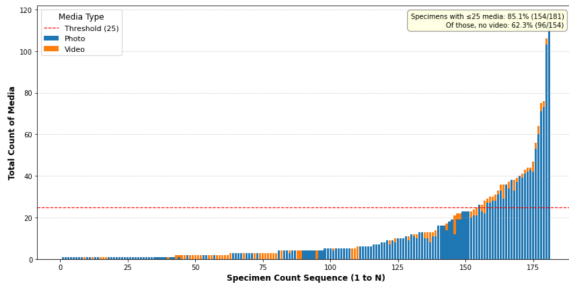
(a) *B. hortorum*(b) *B. lapidarius*(c) *B. pascuorum*(d) *B. pratorum*(e) *B. terrestris*

Figure : Specimens ordered by total media count, showing the imbalance present in the data.

Other Datasets



(a) Jewelwing



(b) Hoverfly



(c) Ladybird



(d) Butterfly

- Worldwide crowdsourced, research-grade observations were curated from iNaturalist.
- Eight species per genus were selected, with ≥ 6 specimens per species and ≥ 4 views per specimen.

Dataset	Species
Jewelwing	<i>C. maculata</i> , <i>C. virgo</i> , <i>C. splendens</i> , <i>C. aequabilis</i> , <i>C. haemorrhoidalis</i> , <i>C. amata</i> , <i>C. angustipennis</i> , <i>C. dimidiata</i>
Hoverfly	<i>E. intricaria</i> , <i>E. tenax</i> , <i>E. arbustorum</i> , <i>E. pertinax</i> , <i>E. horticola</i> , <i>E. interrupta</i> , <i>E. cryptarum</i> , <i>E. rupium</i>
Ladybird	<i>C. septempunctata</i> , <i>C. undecimpunctata</i> , <i>C. transversalis</i> , <i>C. novemnotata</i> , <i>C. trifasciata</i> , <i>C. californica</i> , <i>C. hieroglyphica</i> , <i>C. magnifica</i>
Butterfly	<i>V. cardui</i> , <i>V. atalanta</i> , <i>V. virginienensis</i> , <i>V. annabella</i> , <i>V. itea</i> , <i>V. indica</i> , <i>V. kershawi</i> , <i>V. tameamea</i>

Table : Species composition of the iNaturalist-derived datasets.

Experiments and Protocol

Evaluated Models

Model	Architecture Type
ResNet-101	CNN
EfficientNetV2-S	CNN
Swin-Tiny	Vision Transformer
ConvNeXt-Tiny	Transformer-inspired CNN

Experimental Protocol

- Quantitative Baseline Comparison, ablation, and generalization experiments.
- 5 random specimen-fold seeds; mean results reported.
- 5 specimen-level aggregation rules compared:
 - Majority vote, probability-weighted, confidence-weighted, logit-average, and Top- k vote ($k = 15$).
- Framework parameter: $N = 25$ frames/specimen for BumB-Spec and $N = 6$ for the iNaturalist datasets.

Quantitative Evaluation

Model	Method	Specimen-Level Metrics										Image-Level	
		Prob-weigh.		Logit-Avg.		Majority		Top-k		Conf-weigh.		Metrics	
		F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.	F1	Acc.
ResNet-101 (F1: 68.25, Acc: 86.19)	Group K-Fold	62.05*	87.51	62.56†	87.51	63.17*	87.51	60.65	87.29	60.61*	87.29	54.72*	78.71
	Str. G. K-Fold	72.89*	89.50	71.61*	88.62	70.39*	88.62	72.91*	89.28	71.01†	88.73	62.68	80.00
	Ours (both S1)	72.74*	88.95	73.60†	88.95	72.38*	88.73	73.28*	88.84	73.16*	88.95	63.23	80.15
	Ours (both S2)	61.64†	77.57†	62.85†	77.35†	60.86†	77.13†	61.20†	76.57†	61.17†	77.35†	54.54†	67.49†
	Ours (S1+S2)	74.03†	88.95	73.25*	88.40	72.06*	88.29	73.02*	88.18	73.12*	88.73	59.58*	76.35
EfficientNetV2-S (F1: 70.50, Acc: 88.22)	Group K-Fold	60.39†	87.73	61.86†	88.07	59.39†	87.29	57.84†	87.18	59.61†	87.73	55.68*	80.94
	Str. G. K-Fold	70.94*	90.06	72.54	89.94	70.87*	89.61	70.94*	90.06	71.54*	90.17	63.53	81.25
	Ours (both S1)	75.44*	89.83	76.13*	89.72	75.58*	89.72	75.76*	89.94	76.15*	90.06	64.28*	80.55
	Ours (both S2)	73.36†	85.19†	73.12†	84.64†	71.97†	84.75†	71.63†	83.65†	73.54†	85.30†	60.59†	75.49*
	Ours (S1+S2)	73.95†	89.72	73.58*	89.39	71.68*	88.51	71.68*	88.07	72.91†	89.06	59.22*	77.10
Swin-T (F1: 79.19, Acc: 89.88)	Group K-Fold	81.58*	91.27	81.29*	91.05	77.87*	89.94	81.24*	91.05	81.72†	91.16	68.05	81.98
	Str. Grp. K-Fold	78.27	90.72	77.68	90.50	77.95	90.83	78.86	90.94	78.80	91.16	70.16	83.33
	Ours (both S1)	80.36	90.28	80.64	90.28	79.71	90.17	79.91	89.94	80.82	90.61	68.55	81.38
	Ours (both S2)	76.18†	86.63†	76.61†	86.30†	74.46†	85.86†	78.07†	87.62†	75.82†	86.63†	62.98*	77.21*
	Ours (S1+S2)	80.86	91.16	80.82	90.94	78.93	90.72	80.92	90.39	80.41	90.94	64.35	78.99
ConvNeXt-T (F1: 82.15, Acc: 91.58)	Group K-Fold	82.69*	91.93	81.46*	90.94	81.56	91.60	83.66	92.38	83.13	92.38	65.58	81.53
	Str. G. K-Fold	80.30	92.38	80.20	92.15	79.43	92.04	80.10	92.27	80.90	92.60	68.84	83.14
	Ours (both S1)	82.24	92.27	82.66	92.38	82.16	92.38	81.67	91.82	82.83	92.27	67.41	81.48
	Ours (both S2)	80.17†	87.73†	79.64†	87.40†	79.74†	87.62†	80.45†	88.29†	80.24†	87.96†	65.50	78.96*
	Ours (S1+S2)	86.15	94.03	85.38	93.81	84.93	93.26	85.44	93.48	86.67	94.03	62.86	77.72

*: SD > 3%

†: SD > 5%

Bold: best result per column

- Specimen-level result aggregation consistently outperforms image-level.
- Our framework achieves the best or competitive results against image-level CV strategies in all models.
- Model-wise: ConvNeXt-T > Swin-T > EfficientNetV2-S > ResNet-101

Ablation of Selection Algorithms in Strategy-1

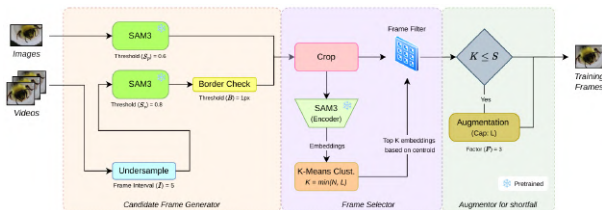
Model	Algorithm	Specimen-Level				Image-Level	
		Prob-weigh.		Majority			
		F1	Acc.	F1	Acc.	F1	Acc.
Swin-T	Random	80.38	90.83	78.86*	90.28	62.49	77.94
	Conf-Weigh.	82.65*	91.82	82.74	91.93	65.21	79.16
	K-Means	80.86	91.16	78.93	90.72	64.35	78.99
	K-Means-T	81.25	91.05	80.94	91.16	64.82	78.72
	FPS	83.19	92.60	80.76	91.38	64.27	78.66
ConvNeXt-T	Random	85.22	93.48	83.47	92.38	61.09	76.60
	Conf-Weigh.	85.80	93.70	85.68	93.37	64.11	78.54
	K-Means	86.15	94.03	84.93	93.26	62.86	77.72
	K-Means-T	82.57*	92.38	82.01	91.82	62.26	77.65
	FPS	85.76	94.25	84.18*	93.37	64.01	79.70

K-Means-T: K-Means after t-SNE reduction

FPS: Farthest Point Sampling.

- **Comparable performance across all selection algorithms**, within 3% Std. Dev. difference suggesting that $N = 25$ images capture sufficient information for most specimens.
- However, Specimen-level aggregation again outperforms image level here.

Ablation of Augmentation Block in Strategy-1



Experiment	Model	Specimen-Level		Image-Level	
		Acc.	F1	Acc.	F1
No Augmentation	ResNet-101	88.95	74.03*	76.35	59.58*
	EfficientNetV2-S	89.72	73.95*	77.10	59.22*
	Swin-Tiny	91.16	80.86	78.99	64.35
	ConvNeXt-Tiny	94.03	86.15	77.72	62.86
Augmentation $F = 3$	ResNet-101	89.94	75.56	76.56	59.44*
	EfficientNetV2-S	90.61	77.47*	77.80	60.36
	Swin-Tiny	91.82	81.31	79.03	62.31*
	ConvNeXt-Tiny	94.48	86.18	78.28	63.03
Augmentation $F = 25$	ResNet-101	89.83	74.39*	77.32	59.02
	EfficientNetV2-S	89.61	76.46*	77.42	59.42
	Swin-Tiny	92.60	83.45	78.59	61.92*
	ConvNeXt-Tiny	93.26	83.85	78.40	62.48

F : number of augmentations generated per selected image. *; SD $> 3\%$

Generalization Experiments

Model	Dataset	Specimen-Level				Image-Level	
		Prob-weigh.		Majority			
		F1	Acc.	F1	Acc.	F1	Acc.
Swin-T	Bumblebee	80.86	91.16	78.93	90.72	64.35	78.99
	Jewelwing	81.22	84.47	78.50	82.26	70.72	75.80
	Butterfly	93.38	96.94	92.35	96.54	88.66	92.95
	Ladybird	92.67	93.21	92.52	93.07	87.47	88.18
	Hoverfly	93.94	94.79	93.47	94.38	88.72	89.56
ConvNeXt-T	Bumblebee	86.15	94.03	84.93	93.26	62.86	77.72
	Jewelwing	81.63	85.24	79.44	83.35	71.03	76.41
	Butterfly	92.77	97.19	91.90	96.52	87.73	92.81
	Ladybird	93.70	94.18	93.05	93.64	87.61	88.42
	Hoverfly	94.28	95.14	93.68	94.60	87.70	89.63

- **Specimen-level aggregation consistently outperforms image-level recognition** for both backbones across all five datasets.
- The improvement is largest for **Bumblebee**, followed by **Jewelwing** > **Hoverfly** > **Ladybird** $\approx$ **Butterfly**.

Qualitative Results

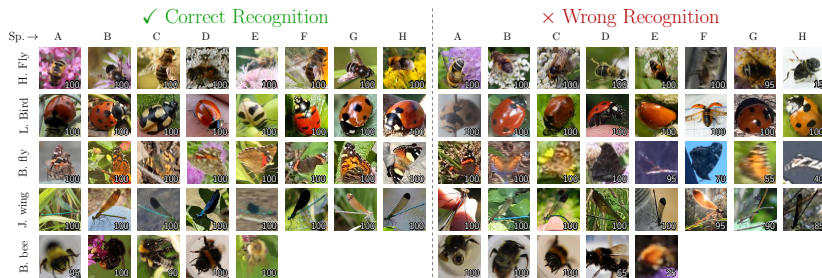


Figure : Top correctly and incorrectly identified specimen per species, with its recognition accuracy (%), across all models and seeds.

Takeaways

- Specimen-level aggregation > image-level evaluation, consistently, across all datasets & backbones.
- S1 and S2 serve different needs: controlled diverse training budget vs. information-rich inference; S1+S2 combined is recommended.
- Different frame-selection algorithm in strategy 1 and specimen aggregation rule perform competitive
- Augmentation of same views for better training does not show superior performance.
- Framework generalizes beyond our proposed bumblebee dataset, which has the highest dataset imbalance/heterogeneity.

Limitations

- Dependence on SAM 3 detections - imperfect on complex backgrounds, sometimes only half individual is localized.
- Border check does not always fully remove partially-visible specimens.
- Videos treated as independent frame collections and no explicit temporal modelling.
- Datasets like jewelwing intra-species variability (sexual dimorphism etc) which is not fully resolved with just visual characteristics.

Possible Future Work

- Use Explainable AI (GradCAM etc) for model interpretability.
- Explore foundation models with PEFT for specimen detection and fine-grained recognition on a bigger scale.
- Scale to larger, more diverse datasets with more locations, seasons, acquisition conditions.
- Developing platform to crowdsource specimen-level images and videos for each insect.

Questions and Reviews

Any Questions?
Comments Please!

Merci! Thank You! धन्यवाद شکریہ